Masking Personally Identifiable Information (PII): Report

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Repository: https://github.com/ikramahelahi/PII-Masking-Assignment

Please find my code, datasets, and results in the following repository.

Dataset and Task

The provided dataset contains tokenized sentences with BIO-style NER labels focused on person names (PER) and email addresses (EMAIL). The task is to detect and mask PII (names and emails) using two approaches:

- A fine-tuned encoder transformer
- A zero-shot Large Language Model (LLM) via prompting.

Approach

- **Fine-tuned model:** An encoder-based transformer was fine-tuned on the prepared training split to predict BIO tags per token. The predicted tags were then used to mask detected PII spans.
- **LLM (zero-shot):** A 1B-parameter LLaMA-style model (*Llama 3.2-1B specifically*) was prompted to extract PII directly from raw text. Outputs were parsed to identify entities without any supervised fine-tuning.

Evaluation Metrics

Performance is evaluated using standard classification metrics, calculated separately for each entity type: **support (count), accuracy, precision, recall, F1-score, false positive rate (FPR)**, and **false negative rate (FNR).** For the LLM, **confusion-matrix-derived counts (TP, FP, FN, TN)** are also included to better understand prediction behavior.

Results

Fine-tuned Transformer (Test Summary)

Metric	Value
Loss	0.0125
Precision	0.9782
Recall	0.9728
F1	0.9755
Accuracy	0.9969

Fine-tuned Transformer: Per-entity Metrics

Entity	Support	Accuracy	Precision	Recall	F1	FPR	FNR
PER	9166	0.996905	0.988468	0.981890	0.985168	0.001340	0.018110
EMAIL	534	0.999989	0.998131	1.000000	0.999065	0.000011	0.000000
PII	9700	0.996893	0.989004	0.982887	0.985936	0.001362	0.017113

Table 1: Per-entity performance for the fine-tuned transformer model.

LLM Zero-shot Masking: Per-entity Confusion and Metrics

Entity	Sup	TP	FP	FN	TN	Acc	Prec	Rec	F1	FPR	FNR
PER	461	300	121	161	4203	0.941066	0.712589	0.650759	0.680272	0.027983	0.349241
EMAIL	29	10	0	19	4756	0.996029	1.000000	0.344828	0.512821	0.000000	0.655172

Table: LLM zero-shot performance with confusion-matrix-derived metrics.

Discussion and Interpretation

The fine-tuned transformer shows strong overall performance, with high accuracy and a good balance between precision and recall for both PER and EMAIL entities. This suggests that supervised training was effective in learning meaningful patterns from the labeled data and that the model generalizes well on unseen examples.

The zero-shot LLM behaves differently. For EMAIL detection, it achieves perfect precision in this run, meaning it does not generate false positives, but its recall is relatively low, so it misses a number of actual email instances. For PER entities, both precision and recall are moderate, indicating that the model can identify names to some extent but is not fully consistent across cases. Overall, this highlights the limitations of relying on prompt-based extraction without task-specific training.

From a practical point of view, the fine-tuned model is more reliable for complete PII masking. The LLM is useful because it is flexible and requires no training, but it tends to miss some entities. Its performance could likely improve with better prompting or more structured output formatting.

Common Errors and Root Causes

- **False negatives (missed entities):** Often arise from unusual name structures, tokenization mismatches, or entities appearing in rare or ambiguous contexts.
- **False positives:** Typically involve short tokens or substrings that resemble names or email patterns.
- **LLM-specific issues:** Prompt ambiguity, limited context handling, and inconsistent output formatting contribute to missed or incorrectly extracted entities.

Recommendations

- Improve LLM performance by adding a few example inputs in the prompt and using a more structured output format such as JSON.
- Improve training data diversity by including more realistic variations of names and email formats to help the model generalize better.
- Combine model predictions with simple rule-based checks, such as regex for emails, to make the system more reliable.
- Perform more detailed evaluation across different types of names (e.g., initials and multi-word names) to better understand model behavior and improve performance.

Conclusion

Both approaches show clear but different strengths. The fine-tuned transformer is a more reliable and high-performing solution for PII masking, while the LLM provides flexibility as a zero-shot baseline without requiring training. For real-world use, the supervised model is the preferred choice, with LLMs acting as supporting tools for quick experimentation or additional signals when needed.